```
restart;
n := 4;
```

$$n := 4 \tag{1}$$

$$result := convert\left(mtaylor\left(u\left(x + p\cdot h,\ t + q\cdot\tau\right),\ \left[h = 0,\ \tau = 0\right],\ n\right),\ diff\right) + O\left(h^4\right) + O\left(\tau^4\right) + O\left(h\cdot\tau^3\right) + O\left(h^2\tau^2\right) + O\left(h^3\cdot\tau\right)$$

$$result := u(x,t) + \left(\frac{\partial}{\partial x}\,u(x,t)\right)p\,h + \left(\frac{\partial}{\partial t}\,u(x,t)\right)q\,\tau + \frac{\left(\frac{\partial^2}{\partial x^2}\,u(x,t)\right)h^2p^2}{2} + \left(\frac{\partial^2}{\partial t\partial x}\,u(x,t)\right)h\,p\,q\,\tau + \frac{\left(\frac{\partial^2}{\partial t^2}\,u(x,t)\right)q^2\tau^2}{2} + \frac{\left(\frac{\partial^3}{\partial x^3}\,u(x,t)\right)h^3p^3}{6} + \frac{\left(\frac{\partial^3}{\partial t\partial x^2}\,u(x,t)\right)h^2p^2q\,\tau}{2} + \frac{\left(\frac{\partial^3}{\partial t^2\partial x}\,u(x,t)\right)h\,p\,q^2\tau^2}{2} + \frac{\left(\frac{\partial^3}{\partial t^3}\,u(x,t)\right)q^3\tau^3}{6} + O\left(h^4\right) + O\left(\tau^4\right) + O\left(h\,\tau^3\right) + O\left(h^2\,\tau^2\right) + O\left(h^3\,\tau\right) \tag{2}$$

$$LU_h := \frac{subs\left(p = 0,\ q = 1,\ result\right) - subs\left(p = 0,\ q = 0,\ result\right)}{\tau} + \frac{c}{2}\cdot\left(\frac{subs\left(p = 0,\ q = 1,\ result\right) - subs\left(p = -1,\ q = 1,\ result\right)}{h} + \frac{subs\left(p = 1,\ q = 0,\ result\right) - subs\left(p = 0,\ q = 0,\ result\right)}{h}\right)$$

$$LU_h := \frac{\left(\frac{\partial}{\partial t}\,u(x,t)\right)\tau + \frac{\left(\frac{\partial^2}{\partial t^2}\,u(x,t)\right)\tau^2}{2} + \frac{\left(\frac{\partial^3}{\partial t^3}\,u(x,t)\right)\tau^3}{6}}{\tau} + \frac{1}{2}\left(c\left(\frac{1}{h}\left(\left(\frac{\partial}{\partial x}\,u(x,t)\right)h - \frac{\left(\frac{\partial^2}{\partial x^2}\,u(x,t)\right)h^2}{2} + \left(\frac{\partial^2}{\partial t\partial x}\,u(x,t)\right)h\,\tau + \frac{\left(\frac{\partial^3}{\partial x^3}\,u(x,t)\right)h^3}{6} - \frac{\left(\frac{\partial^3}{\partial t\partial x^2}\,u(x,t)\right)h^2\tau}{2} + \frac{\left(\frac{\partial^3}{\partial t^2\partial x}\,u(x,t)\right)h\,\tau^2}{2}\right)\right.\right. \tag{3}$$

$$+\frac{\left(\frac{\partial}{\partial x}\,u(x,t)\right)h+\frac{\left(\frac{\partial^2}{\partial x^2}\,u(x,t)\right)h^2}{2}+\frac{\left(\frac{\partial^3}{\partial x^3}\,u(x,t)\right)h^3}{6}}{h}\Bigg)\Bigg)$$

$f_h := diff(u(x,t),t) + c\cdot diff(u(x,t),x);$

$$f_h := \frac{\partial}{\partial t}\,u(x,t) + c\left(\frac{\partial}{\partial x}\,u(x,t)\right) \tag{4}$$

$\xi := simplify(LU_h) - f_h;$

$$\xi := \frac{\left(\frac{\partial^3}{\partial x^3}\,u(x,t)\right)c\,h^2}{6} - \frac{\left(\frac{\partial^3}{\partial t\partial x^2}\,u(x,t)\right)c\,h\,\tau}{4} + \frac{\left(\frac{\partial^3}{\partial t^2\partial x}\,u(x,t)\right)c\,\tau^2}{4} \tag{5}$$
$$+\frac{\left(\frac{\partial^2}{\partial t\partial x}\,u(x,t)\right)c\,\tau}{2}+\frac{\left(\frac{\partial^3}{\partial t^3}\,u(x,t)\right)\tau^2}{6}+\frac{\left(\frac{\partial^2}{\partial t^2}\,u(x,t)\right)\tau}{2}$$

$\xi + O(h^3) + O(\tau^3)$

$$\frac{\left(\frac{\partial^3}{\partial x^3}\,u(x,t)\right)c\,h^2}{6} - \frac{\left(\frac{\partial^3}{\partial t\partial x^2}\,u(x,t)\right)c\,h\,\tau}{4} + \frac{\left(\frac{\partial^3}{\partial t^2\partial x}\,u(x,t)\right)c\,\tau^2}{4} + \frac{\left(\frac{\partial^2}{\partial t\partial x}\,u(x,t)\right)c\,\tau}{2} \tag{6}$$
$$+\frac{\left(\frac{\partial^3}{\partial t^3}\,u(x,t)\right)\tau^2}{6}+\frac{\left(\frac{\partial^2}{\partial t^2}\,u(x,t)\right)\tau}{2}+O(h^3)+O(\tau^3)$$

$\xi := collect(\xi, [\mathrm{tau}, h], simplify):$

$\xi + O(h^3) + O(\tau^3)$

$$\left(\frac{\left(\frac{\partial^3}{\partial t^2\partial x}\,u(x,t)\right)c}{4}+\frac{\frac{\partial^3}{\partial t^3}\,u(x,t)}{6}\right)\tau^2+\left(-\frac{\left(\frac{\partial^3}{\partial t\partial x^2}\,u(x,t)\right)c\,h}{4}+\frac{\left(\frac{\partial^2}{\partial t\partial x}\,u(x,t)\right)c}{2}\right. \tag{7}$$
$$\left.+\frac{\frac{\partial^2}{\partial t^2}\,u(x,t)}{2}\right)\tau+\frac{\left(\frac{\partial^3}{\partial x^3}\,u(x,t)\right)c\,h^2}{6}+O(h^3)+O(\tau^3)$$

$new_\xi := subs($
$diff(u(x,t),t\$3) = -c^3\cdot diff(u(x,t),x\$3),$
$diff(u(x,t),t\$2,x) = c^2\cdot diff(u(x,t),x\$3),$
$diff(u(x,t),t,x\$2) = -c\cdot diff(u(x,t),x\$3),$
$diff(u(x,t),t\$2) = c^2\cdot diff(u(x,t),x\$2),$
$diff(u(x,t),x,t) = -c\cdot diff(u(x,t),x\$2),$

$diff(u(x,t),t) = -c \cdot diff(u(x,t),x),$

ξ

$):$

$$new_\xi + O(h^3) + O(\tau^3)$$

$$\frac{c^3 \left(\frac{\partial^3}{\partial x^3} u(x,t)\right) \tau^2}{12} + \frac{\left(\frac{\partial^3}{\partial x^3} u(x,t)\right) c^2 h \tau}{4} + \frac{\left(\frac{\partial^3}{\partial x^3} u(x,t)\right) c h^2}{6} + O(h^3) + O(\tau^3) \quad (8)$$

$\#Невязка = O(\tau^2) + O(\tau \cdot h) + O(h^2); |h \cdot \tau| \le \frac{h^2 + \tau^2}{2}$, *поэтому невязка* $= O(\tau^2) + O(h^2)$

. Значит :

$k := 2; m := 2;$

$$k := 2$$

$$m := 2 \quad (9)$$

#Предельный переход $\tau = \varkappa \cdot h$

$pr_1 := collect(subs(\tau = \varkappa \cdot h, new_\xi), [h], simplify)$

$$pr_1 := \frac{\left(c^3 \varkappa^2 + 3 \varkappa c^2 + 2 c\right) \left(\frac{\partial^3}{\partial x^3} u(x,t)\right) h^2}{12} \quad (10)$$

#Порядок апроксимации p=2

$C_1 = coeff(pr_1, h, 2)$

$$C_1 = \frac{\left(c^3 \varkappa^2 + 3 \varkappa c^2 + 2 c\right) \left(\frac{\partial^3}{\partial x^3} u(x,t)\right)}{12} \quad (11)$$

#Предельный переход $\tau = h^2 \backslash \mu$

$pr_2 := collect\left(subs\left(\tau = \frac{h^2}{\mu}, new_\xi\right), [h], simplify\right)$

$$pr_2 := \frac{c^3 \left(\frac{\partial^3}{\partial x^3} u(x,t)\right) h^4}{12 \mu^2} + \frac{\left(\frac{\partial^3}{\partial x^3} u(x,t)\right) c^2 h^3}{4 \mu} + \frac{\left(\frac{\partial^3}{\partial x^3} u(x,t)\right) c h^2}{6} \quad (12)$$

#Порядок апроксимации тоже p=2, но C_1 будет другая:

$C_1 = coeff(pr_2, h, 2)$

$$C_1 = \frac{\left(\frac{\partial^3}{\partial x^3} u(x,t)\right) c}{6} \quad (13)$$

#Г-форма первого дифференциального приближения

$g_1 := -\left(coeff\left(coeff\left(\xi, \text{tau}, 1\right), h, 0\right)\cdot\tau + coeff\left(coeff\left(\xi, \text{tau}, 0\right), h, 1\right)\cdot h\right)$:

$f_h = \text{g_1}$

$$\frac{\partial}{\partial t}\,u(x,t) + c\left(\frac{\partial}{\partial x}\,u(x,t)\right) = -\left(\frac{\left(\frac{\partial^2}{\partial t\partial x}\,u(x,t)\right)c}{2} + \frac{\frac{\partial^2}{\partial t^2}\,u(x,t)}{2}\right)\tau \tag{14}$$

#П-форма первого дифференциального приближения

$p_1 := subs\left(diff\left(u(x,t), t\$2\right) = c^2\cdot diff\left(u(x,t), x\$2\right), c\cdot diff\left(u(x,t), t, x\right) = -c^2\cdot diff\left(u(x,t), x\$2\right), g_1\right)$

$$p_1 := 0 \tag{15}$$

$f_h = p_1$

$$\frac{\partial}{\partial t}\,u(x,t) + c\left(\frac{\partial}{\partial x}\,u(x,t)\right) = 0 \tag{16}$$

#Г-форма второго дифференциального приближения

$g_2 := -\left(coeff\left(coeff\left(\xi, \text{tau}, 2\right), h, 0\right)\cdot\tau^2 + coeff\left(coeff\left(\xi, \text{tau}, 0\right), h, 2\right)\cdot h^2 + coeff\left(coeff\left(\xi, \text{tau}, 1\right), h, 1\right)\cdot h\cdot\tau\right)$:

$f_h = \text{g_2}$

$$\frac{\partial}{\partial t}\,u(x,t) + c\left(\frac{\partial}{\partial x}\,u(x,t)\right) = -\left(\frac{\left(\frac{\partial^3}{\partial t^2\partial x}\,u(x,t)\right)c}{4} + \frac{\frac{\partial^3}{\partial t^3}\,u(x,t)}{6}\right)\tau^2 - \frac{\left(\frac{\partial^3}{\partial x^3}\,u(x,t)\right)c\,h^2}{6} + \frac{\left(\frac{\partial^3}{\partial t\partial x^2}\,u(x,t)\right)c\,h\,\tau}{4} \tag{17}$$

#П-форма второго дифференциального приближения

$p_2 := new_\xi$

$$p_2 := \frac{c^3\left(\frac{\partial^3}{\partial x^3}\,u(x,t)\right)\tau^2}{12} + \frac{\left(\frac{\partial^3}{\partial x^3}\,u(x,t)\right)c^2\,h\,\tau}{4} + \frac{\left(\frac{\partial^3}{\partial x^3}\,u(x,t)\right)c\,h^2}{6} \tag{18}$$

$p_2 := simplify\left(subs\left(\tau = \frac{\gamma\cdot h}{c}, p_2\right)\right)$

$$p_2 := \frac{\left(\frac{\partial^3}{\partial x^3}\,u(x,t)\right)c\,h^2\left(\gamma+2\right)\left(\gamma+1\right)}{12} \tag{19}$$

$f_h = -p_2$

$$\frac{\partial}{\partial t} u(x,t) + c\left(\frac{\partial}{\partial x} u(x,t)\right) = -\frac{\left(\frac{\partial^3}{\partial x^3} u(x,t)\right) c\, h^2 (\gamma+2)(\gamma+1)}{12} \tag{20}$$